ST. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) • OMR, Chennai - 119

LAB RECORD

Professional English Lab Integrated

CLASS CODE: EN25C09

STUDENT DETAILS

Student Name: Harshini Thiyagarajan Logeshwari

Roll No: 26CS217

Date:

Subject Name: Professional English

HARSHINI THIYAGARAJAN LOGESHWARI

BE Computer Science and Engineering | Aspiring Cybersecurity Analyst

Links	**GitHub:** https://github.com/tlharshini123-source
Summary	I am a curious and adaptable Computer Science student with a strong interest in cybersecurity and computer networks. I enjoy learning new technologies, exploring how systems work, and improving my coding and problem-solving skills. My goal is to build a strong foundation in cybersecurity and grow into a skilled Cybersecurity Analyst.
Introduction	Hi, I'm Harshini, a first-year BE Computer Science and Engineering student at St. Joseph's College of Engineering. I am interested in networking and cybersecurity, and I enjoy learning new concepts, coding, and taking on challenges. I am an enthusiastic team player who values adaptability, leadership, and continuous learning. I look forward to exploring hackathons and practical projects to strengthen my technical knowledge.
Skills	**Programming:** Python, C (beginner) **Database:** MySQL (foundational knowledge) **Technical interests:** Computer Networking, Cybersecurity fundamentals **Strengths:** Adaptability, Leadership, Problem-solving, Quick communication
Education	**BE – Computer Science and Engineering** St. Joseph's College of Engineering, OMR, Chennai First Year | 2026–2030
Career Goal	To develop strong skills in cybersecurity and networking, gain practical experience through projects and hackathons, and pursue a career as a Cybersecurity Analyst.
Areas of Interest	Cybersecurity • Computer Networking • Ethical Hacking (learning) • Programming • Technology exploration

Impressive Personality: Biographical Note

Mikko Hyppönen – Finnish Cybersecurity Expert, Researcher and Author

Mikko Hyppönen is a well-known Finnish cybersecurity expert who has spent more than three decades studying computer viruses, malware and online threats. He is respected internationally for his work in protecting computer systems and explaining complex security issues to the public. He began his security research career in 1991 and later became one of the best-known voices in the field of computer security. In 2025, he moved to a new role as Chief Research Officer at Sensofusion, a Finnish technology company working on drone-defence systems. [1][2]

Early Life and Interest in Computers

Hyppönen's interest in computers began when he was young. His mother worked at Finland's state computing centre and sometimes brought home computer punch cards and paper tape. These materials introduced him to the world of computing before personal computers became common. He spent time exploring them and gradually developed a strong interest in technology. In a later interview, he explained that this early exposure helped shape his interest in computers and security. [3]

Rather than treating technology only as a source of entertainment, Hyppönen became interested in understanding how computer systems worked and how they could be misused. This curiosity eventually led him towards computer security research, a field that was still developing rapidly in the early 1990s. His career shows how curiosity and continuous learning can grow into professional expertise.

Beginning of His Career

In 1991, while studying computer science, Hyppönen joined a small Finnish company called Data Fellows as one of its early employees. His work involved examining computer viruses and understanding how they operated. Data Fellows later became known as F-Secure, a company that developed security software. Over the years, Hyppönen took on greater responsibilities and became a leading researcher in the organisation. He worked in the security industry for 34 years, contributing to the study and defence of computers against malicious software. [2] [4]

Impressive Personality: Biographical Note

Mikko Hyppönen | Professional English – I (Lab Integrated)

Work in Malware Research and Cybersecurity

A major part of Hyppönen's career focused on malware research. Malware is software designed to harm devices, steal information or gain unauthorised access to computer systems. By studying how malicious programs spread and behave, security researchers can develop ways to detect and prevent them. Hyppönen led research teams during several major malware outbreaks and worked to help people understand the changing nature of cybercrime. His work also involved cooperating with law-enforcement agencies on cybercrime investigations. [5] He became known for communicating technical subjects in a clear and understandable way. He has written about cybersecurity for international publications, including The New York Times, Wired and Scientific American. He has also spoken at major security events such as DEF CON, RSA and Black Hat, and has lectured at universities including Oxford, Harvard and MIT. These activities helped him share knowledge with students, researchers, technology professionals and the general public. [1][4]

Books, Talks and Public Awareness

Hyppönen is also an international speaker and author. His book *If It's Smart, It's Vulnerable* discusses the security risks created by internet-connected devices. The central idea is that when more everyday objects become connected to the internet, they can also become targets for cyberattacks. This message is especially important as smart phones, televisions, home devices and other connected technologies become part of daily life.
[3][6]

Through his articles, interviews and public talks, he encourages people to take digital security seriously. His work highlights the importance of software updates, careful online behaviour, privacy and awareness of cyber threats. He demonstrates that cybersecurity is not only a technical subject; it is also connected to the safety and privacy of ordinary people.

Recognition and Influence

Hyppönen has received international recognition for his contribution to computer security. He has been included in lists recognising influential people in technology and global thought, and he has been invited to speak at leading conferences around the world. His influence comes not only from technical research but also from his ability to explain why cybersecurity matters to society. [1][5]

Impressive Personality: Biographical Note

Mikko Hyppönen | Professional English – I (Lab Integrated)

A New Chapter in His Career

In June 2025, Hyppönen announced that he was moving from traditional cybersecurity work to a new challenge at Sensofusion, a Finnish company specialising in anti-drone technology. He took up the role of Chief Research Officer. His experience in detecting hidden digital threats is relevant to this work because both cybersecurity and drone defence involve identifying risks that may change their behaviour to avoid detection. He has said that he will continue to speak about cybersecurity and technology. [2][7]

Personal Qualities and Lessons for Students

Mikko Hyppönen's career reflects curiosity, dedication, adaptability and a commitment to lifelong learning. He entered a developing area of technology, continued learning as threats changed and shared his knowledge with people around the world. His career also shows the value of communication: a cybersecurity professional must be able to investigate technical problems and explain their importance clearly.

For students of Computer Science and Engineering, his journey is an example of how an interest in computers can develop into a meaningful career. Students can learn from his habit of exploring new technologies, strengthening their fundamentals and staying aware of changes in the digital world. His work encourages future professionals to use their technical knowledge responsibly and help make technology safer for everyone.

Conclusion

Mikko Hyppönen is an inspiring personality in the field of cybersecurity. From his early interest in computers to decades of malware research, international speaking and authorship, he has made an important contribution to digital security awareness. His move into drone-defence research in 2025 also demonstrates his willingness to take on new challenges. His life and work remind us that knowledge, curiosity and continuous learning are essential for success in the fast-changing world of technology.

References

[1] RSA Conference – Mikko Hyppönen: https://www.rsaconference.com/experts/mikko-hypponen

[2] SecurityWeek – Mikko Hypponen joins Sensofusion (3 June 2025): https://www.securityweek.com/mikko-hypponen-joins-anti-drone-company-sensofusion/

[3] Baillie Gifford – Mikko Hyppönen: a veteran's take on the future of cybersecurity: https://www.bailliegifford.com/en/uk/individual-investors/insights/ic-article/2023-q1-author-interview-mikko-hypponen-10019299/ [4] Mikko Hyppönen – Official website: https://mikko.com/about

[5] Lifeboat Foundation – Mikko H. Hyppönen: https://lifeboat.com/bios.mikko.h.hypponen.shtml

[6] F-Secure – Staying ahead of cyber threats in the age of AI: https://assets.f-secure.com/p/staying-ahead-of-cyber-threats-in-the-age-of-ai.pdf

[7] The Register – Hyppönen moves from infosec to drone defence (4 June 2025): https://www.theregister.com/security/2025/06/04/mikko_hyppoenen_pivots_to_drones_inspired_by_war/1048036

Summary of Lecture Notes: Venom Studies & Toxinology

NPTEL: Fundamentals of Toxinology, Venoms and Therapeutics (IIT Madras & IISc)

Week 1: Introduction to Toxinology and Venomous Organisms

Toxinology is the specialized branch of science that studies biological toxins produced by plants, animals, and microorganisms. Unlike poison, which is typically ingested or absorbed passively, venom is a specialized biological secretion actively delivered via apparatuses such as fangs, stingers, or spurs to facilitate predation, defense, or competition.

The evolutionary trajectory of venom systems shows convergence across diverse taxa. Key concepts include the distinction between biological function (subduing prey versus deterring predators) and chemical composition, which ranges from simple small molecules to complex enzymatic proteins.

Week 2: Snake Venoms – Composition and Pathophysiology

Snake venoms are complex mixtures primarily composed of enzymatic and non-enzymatic proteins. The major toxic families include:

- **Neurotoxins:** Target neuromuscular junctions, inducing flaccid paralysis by blocking acetylcholine receptors.
- **Hemotoxins & Phospholipases A2:** Disrupt cardiovascular stability, cause systemic hemorrhage, and trigger severe local tissue necrosis.
- **Cytotoxins:** Destroy cell membranes, leading to rapid localized tissue destruction and necrosis.

Week 3: Invertebrate Toxins – Arachnids, Insects, and Marine Species

Arachnid and insect venoms utilize small peptide toxins engineered for targeted ion channel modulation. Spider venoms (e.g., latrotoxins) induce massive neurotransmitter release, while scorpion venoms block voltage-gated potassium and sodium channels, producing acute autonomic excitation and severe pain.

Marine invertebrates like cone snails (Conus species) produce conopeptides—highly specific neurotoxic peptides that target voltage-gated ion channels and GPCRs with extraordinary potency, serving as vital lead compounds in drug discovery.

Week 4: Molecular Mechanisms of Venom Action

Venom components exert toxicity through precise molecular interactions with physiological receptors, ion channels, and enzymatic substrates. Key biochemical mechanisms include competitive antagonism at postsynaptic receptors, enzymatic degradation of extracellular matrix proteins (e.g., hyaluronidase activity accelerating tissue spread), and activation of prothrombin triggering disseminated intravascular coagulation (DIC).

Week 5: Antivenom Production and Immunotherapy

Antivenoms remain the standard specific therapy for envenomation. Production involves hyperimmunizing donor host animals (typically horses or sheep) with sub-lethal, gradually increasing doses of targeted venoms. Plasma harvested from immunized animals undergoes enzymatic digestion (using pepsin or papain) to yield purified divalent F(ab')2 or monovalent Fab antibody fragments.

Major challenges in antivenom therapy include batch-to-batch variation, low paraspecific neutralization efficiency across geographic variants, and immediate severe adverse reactions such as anaphylaxis or delayed serum sickness.

Week 6: Venomics, Proteomics, and Next-Generation Antivenoms

Modern venomics integrates high-throughput proteomics, transcriptomics, and bio-informatics to map complete venom profiles. This multi-omic approach enables precise identification of lowabundance toxic peptides and geographical intraspecific variations in venom composition.

Next-generation therapeutics focus on recombinant monoclonal antibody cocktails, synthetic smallmolecule inhibitors (e.g., metalloproteinase and phospholipase inhibitors), and universal antivenom platforms designed to bypass traditional animal-based immunization protocols.

Week 7: Therapeutics and Drug Discovery from Venoms

Despite their lethality, venom components offer unparalleled structural specificity for drug development. Successful venom-derived pharmaceuticals include:

- **Captopril:** Synthetic derivative of peptides from *Bothrops jararaca* venom, revolutionized treatment of hypertension (ACE inhibition).
- **Eptifibatide & Tirofiban:** Antiplatelet agents modeled after viperid disintegrins to mitigate acute cardiac ischemic events.
- **Ziconotide:** A synthetic peptide derived from *Conus magus* cone snail venom, acting as a nonopioid analgesic for chronic severe pain.

Week 8: Clinical Management, First Aid, and Ecological Perspectives

Effective clinical management of envenomation relies on rapid diagnostic assessment, systemic stabilization, and prompt, targeted antivenom administration. Outdated first-aid methods (such as incision, suction, or tight arterial tourniquets) are strongly discouraged due to increased risk of localized necrosis and severe tissue damage.

From an ecological perspective, venomous species play critical roles in ecosystem stability as apex and mesopredators. Conserving venomous taxa is essential not only for biodiversity, but also for preserving unexplored bio-molecular reservoirs vital for future medicinal breakthroughs.

SWOC Presentation

Strategic Career Roadmap: SWOC Analysis (slides)

Strategic Career Roadmap: SWOC Analysis

Path to Principal Cloud Architect / Director of Engineering

Candidate: Inba Sakkan | 1st Year B.E./B.Tech CSE | Batch 2026 – 2030

Goal: get deep into cloud and systems early, then move toward architecture and engineering leadership.

Executive Summary & Career Vision

1 Vision

Start as a first-year CSE student and grow into a Principal Cloud Architect/ Director of Engineering in the leadership mode.

2 Objective

Learn Linux and networking properly, then cloud. Put design, judgment and problem-solving ahead of routine coding.

3 Framework

Use a SWOC matrix (Strengths, Weaknesses, Opportunities, Challenges) to draw up a 4-year milestone roadmap.

SWOC Matrix at a Glance



Strengths

1	Excellent problem solver	I can sit down and think a complex problem through. That is the core skill of architecture.
2	Linux enthusiast	I'm comfortable working terminal-first, and I like finding out how systems work.
3	Python foundation	Hands-on practice through two personal projects, Rover-Bot and sattu.
4	Public portfolio started	An active GitHub profile with a README, so recruiters can see the projects.

Weaknesses

Working style

Impatience. I tend to rush into a problem and hit a brick wall before I've planned anything.

It's the habit most worth fixing. Architects earn their keep by thinking before they build.

Experience gaps

I'm a beginner in operating systems, networks and databases, none of which I've studied formally yet.

No production experience, internship or certification so far.

Opportunities

AI raises the coding baseline

More and more routine boilerplate gets automated, so value moves toward architecture, systems design and decision-making.

Growing cloud demand

Migration, DevOps, platform engineering and FinOps roles keep growing.

Student programs

AWS Educate, the GitHub Student Developer Pack and Azure for Students all offer free credits.

Certifications and community

Entry-level certs such as AWS Cloud Practitioner and AZ-900, plus open source, hackathons and meetups.

Challenges

Overstimulating environments

Noisy, high-energy settings overstimulate me. That lists labs, hackathons and open-plan offices.

Shrinking junior pipeline

That came AI shift means fewer boilerplate junior roles, so I need to show design ability early.

Experience paradox

Entry roles often ask for experience I haven't had a chance to get yet.

Tooling churn and cloud cost

Tools change fast, and personal labs risk surprise bills.

Action Plan: Closing the Gaps

1 Impatience

Before any project, I write a short design note (goal, components, risks) and sketch a diagram. Only then do I build.

2 Knowledge gap

Use Linux every day. Study networking basics (TCP/IP, DNS, subnetting) alongside coursework.

3 Credential gap

Take AWS Cloud Practitioner before entering Year 3.

Action Plan: Using Opportunities

Use AI as a tool

Let it write the boilerplate. I focus on design, review and debugging, and I should be able to explain every line I ship.

Manage noise

Book deep-work blocks for quiet hours, wear headphones, and choose seats or small group formats where possible.

Build production-style projects

CI/CD, IaC and containerised apps, each with an architecture diagram in its README.

Stay cost-safe

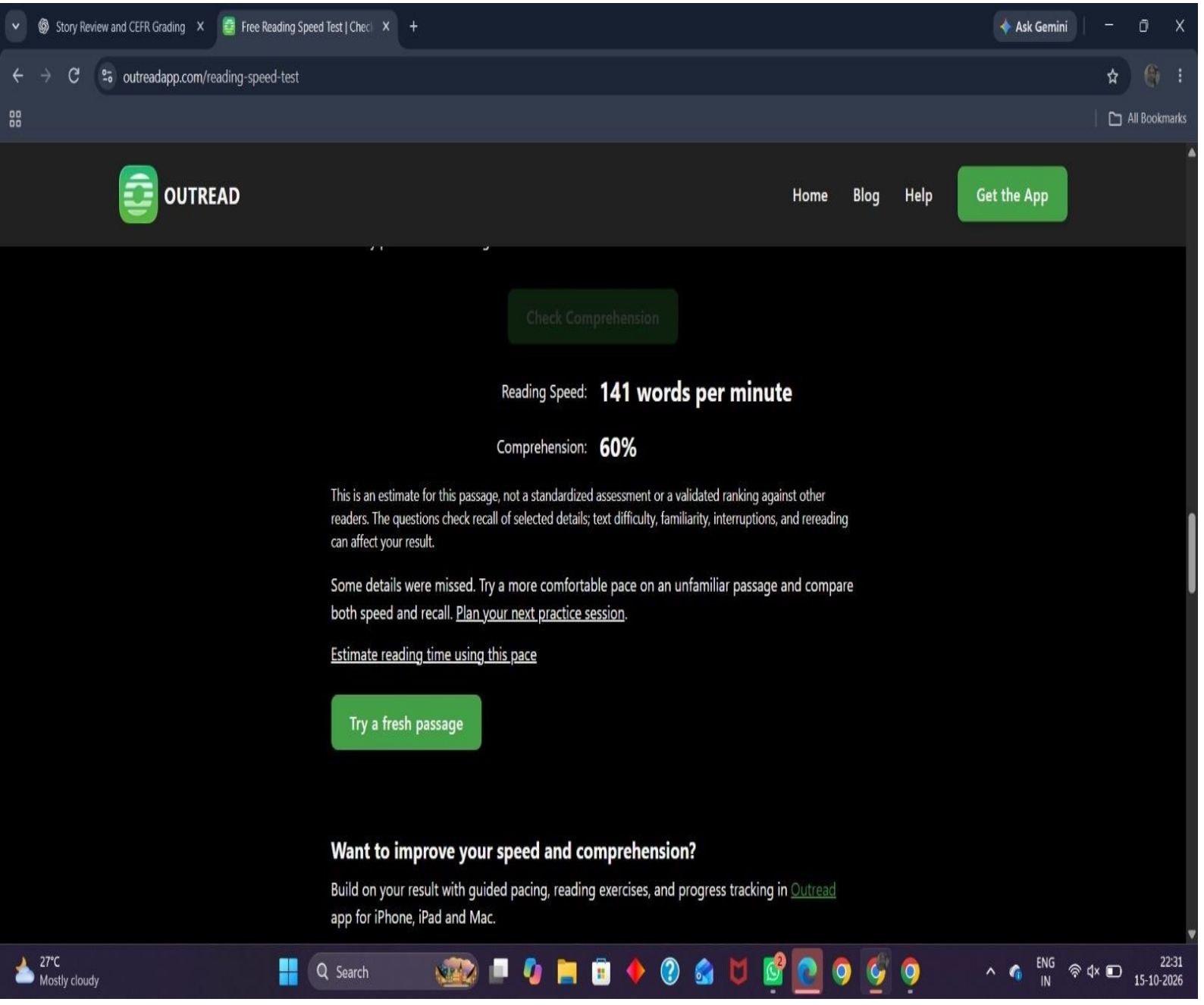
Set billing alarms. Run every cloud lab on student credits.

4-Year Milestones Matrix

Year	Primary Focus	Target Deliverable	Target Credential
Year 1 (2026–27)	Linux, networking, cloud basics	Portfolio site on AWS S3 / CloudFront	AWS Cloud Practitioner
Year 2 (2027–28)	Docker, Python automation, IaC	3-tier VPC deployed with Terraform	AWS Solutions Architect Associate
Year 3 (2028–29)	Kubernetes, CI/CD, DevSecOps	Microservices on EKS via GitHub Actions	CKA or Terraform Associate
Year 4 (2029–30)	SRE, FinOps, leadership	Multi-cloud disaster recovery capstone, lead a team project	Placement / off-campus hiring


Conclusion & Key Takeaways
1
Think first, then build
Planning before coding is what turns my problem-solving strength into architect-level work.
2
Fundamentals plus AI
AI-generated code is only trustworthy when someone understands the systems underneath it.
3
Verdict
If I plan with discipline and keep doing hands-on cloud work, I should be able to build a profile aimed at architecture and engineering leadership.

SPEAK AND IMPROVE



COMMON EUROPEAN FRAMEWORK OF REFERENCE FOR LANGUAGES

Language Proficiency Score Card

OVERALL LEVEL
B2
VANTAGE

LEARNER HARSHINI THIYAGARAJAN LOGESHWARI

LANGUAGE English [EN25C09]

DATE 27-10-2026

A1	A2	B1	**B2**	C1	C2
BASIC USER		INDEPENDENT USER		PROFICIENT USER	

Can interact with fluency and spontaneity, produce clear, detailed text on a wide range of subjects, and explain a viewpoint on a topical issue.

Professional English Practice Worksheets – Solved

Course Code: EN25C09 | Subject: Professional English – I (Lab Integrated) | Worksheet 1: Model Exam I (Parts A, B, C), followed by Worksheet 2: Model – I Practice Worksheet.

Worksheet 1 (EN25C09 Model Exam I) – Part A (15 × 1 = 15 Marks)

1. **Answer:** monitors
2. **Answer:** recorded
3. **Answer:** stronger
4. **Answer:** must 5. **Answer:** can
6. **Answer:** The laboratory developed a new testing device that measures material strength with high precision.
7. **Answer:** rises
8. **Answer:** would not have failed
9. **Answer:** The mechanics installed dual damper springs to isolate chassis resonance.
10. **Answer:** optimizing
11. **Answer:** 50 mL of hydrochloric acid was poured into the titration flask.
12. **Answer:** smoothly
13. **Answer:** The project supervisor asked the engineer whether he had calibrated the pressure sensor before testing.
14. **Answer:** The incorrect noun form is "equipments", because "equipment" is uncountable. Correct sentence: The laboratory purchased several new pieces of equipment for conducting material testing.
15. **Answer:** worse

Worksheet 1 – Part B (10 × 2 = 20 Marks)

1. **Answer:** The engineer calibrated the pressure sensor and verified its accuracy.
2. **Answer (Portability):** A laptop is more portable than a desktop computer because it is lighter and can be carried anywhere.

   Answer (Screen Size): A desktop computer usually has a larger screen than a laptop, so it is more comfortable for long hours of detailed work.
3. **Answer:** The engineer designed a new cooling system, which reduces the temperature of industrial machines.
4. (a) **Answer:** If the battery voltage drops below the required level, the device will shut down automatically.

   (b) **Answer:** If the technician had checked the connections earlier, the fault would have been detected before the breakdown.
5. **Answer:** The solution was heated to 80°C before testing.
6. **Answer:** The project manager asked the technician whether he had inspected the pressure valves before starting the compressor.
7. (a) **Answer:** The engineer received a lot of feedback on the new design. ("Feedback" is uncountable, so it cannot take "many" or the plural form "feedbacks".)

   (b) **Answer:** The students received several pieces of useful advice from their teacher. ("Advice" is uncountable, so "advices" is wrong.)
8. (a) **Answer:** The technician wears safety gloves to protect his hands.

   (b) **Answer:** The cooling fan operates continuously to prevent the motor from overheating.
9. (a) **Answer:** efficiently

   (b) **Answer:** carefully

10. (a) **Answer:** can

(b) **Answer:** must not

Worksheet 1 – Part C (5 × 13 = 65 Marks)

Question 26 (a) Safety instructions for safe and responsible use of the internet

1. Use a strong, unique password of at least twelve characters, mixing letters, numbers and symbols, for everyaccount.
2. Enable two-factor authentication on your email, banking and social media accounts.
3. Do not click on links or open attachments from unknown or suspicious senders.
4. Never share personal details such as OTPs, passwords, Aadhaar or bank details with anyone online.
5. Check that a website address begins with "https" and shows a padlock before entering any sensitiveinformation.
6. Keep your operating system, browser and antivirus software updated at all times.
7. Avoid banking or shopping on public Wi-Fi; use a trusted VPN if you must connect.
8. Report cyberbullying, online fraud and phishing attempts immediately to a trusted adult, the platform, or theNational Cyber Crime Helpline (1930).

Question 27 (a) Descriptive essay: A traditional art form in my region and how modern initiatives can preserve it

Selected Question: 27.a

Bharatanatyam: The Dance That Carries Tamil Nadu's Soul

Among the many traditional art forms of Tamil Nadu, Bharatanatyam is the one that has carried our culture across the centuries. It began inside the temples, where devadasis performed it as an offering to the deity. The epic Silappathikaram describes classical dance in detail, and the carvings on the Brihadeeswarar Temple at Thanjavur and the Chidambaram Nataraja Temple show its postures in stone. In the nineteenth century, the Thanjavur Quartet organised the repertoire into the form we see today.

Traditionally, a dancer trains for many years under a guru. A performance blends nritta (pure rhythm), nritya (expressive storytelling) and natya (drama), told through hand gestures called mudras, facial expressions and Carnatic music. The dancer wears a bright silk costume, temple jewellery and ankle bells called salangai. The first full solo performance, the arangetram, marks the end of a student's formal training.

Today, many young people feel the art form is old-fashioned and too expensive to learn. Modern initiatives can change this. Schools and colleges can include classical dance in cultural clubs and the curriculum, as institutions like Kalakshetra in Chennai already do. Online classes, recorded performances and social media reels can reach learners in small towns. Festivals such as the Margazhi season in Chennai give young artists a stage, and scholarships can help families who cannot afford training.

Bharatanatyam is more than entertainment; it is a living record of our history, music and devotion. If we support it through education, technology and public platforms, the next generation will carry the dancing bells forward.

Question 28 (b) Formal letter of request to the Principal

Selected Question: 28.b

Harshini Thiyagarajan Logeshwari
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Principal
St. Joseph's College of Engineering

OMR, Chennai – 600 119

Subject: Request for permission to attend a technical workshop on Artificial Intelligence and Emerging Technologies

Respected Sir/Madam,

I am a first-year B.E. Computer Science and Engineering student of your college. I have come to know that [Name of Institution] is organising a one-day technical workshop on Artificial Intelligence and Emerging Technologies on [date]. I request your kind permission to attend this workshop.

My objective in attending is to learn how artificial intelligence is applied in real engineering problems and to gain hands-on exposure to tools that are not covered in detail in our first-year syllabus. I also hope to meet experts and students from other institutions and learn from their experience.

I expect to learn the basics of machine learning, practical uses of AI in software and cloud systems, and the latest trends in emerging technologies. After the workshop, I will share the notes and key points with my classmates and use what I learn in my Python projects.

I request you to kindly grant me on-duty leave for the day of the workshop and permit me to travel to the venue. I will make up for any classes missed and will submit the participation certificate to the department after the workshop.

I would be grateful if you could grant me this permission.

Thank you for your kind consideration.

Yours obediently,
Harshini Thiyagarajan Logeshwari
(Register No.: [Register No.])

Question 29 (a) Analytical essay: AI-based assessment versus traditional teacher-based assessment

Selected Question: 29.a

AI-Based Assessment versus Traditional Teacher-Based Assessment in Engineering Education

Assessment decides how students learn, so the choice between AI-based and teacher-based methods matters in engineering education. Each has clear strengths and weaknesses.

In terms of accuracy, AI tools are consistent. An automated system can evaluate hundreds of code submissions or objective answers using the same rules every time, without getting tired. Teachers, on the other hand, can understand a student's reasoning in a long derivation or a design report, which software often misses. Therefore, AI is accurate for objective answers, whereas teachers are better for open-ended ones.

Feedback time is where AI wins most clearly. Students receive results and error explanations within seconds, while manual evaluation of a large class may take weeks. Quick feedback lets students correct their mistakes while the topic is still fresh. Conversely, a teacher's comments arrive more slowly but are usually more personal and motivating.

Regarding fairness, AI does not show favouritism or mood-based marking. However, it can carry hidden bias from the data it was trained on, and it may penalise students who give unusual but valid solutions. A teacher can notice such cases and adjust the marks.

Finally, human judgement remains important. Engineering involves creativity, ethics and teamwork, which cannot be reduced to a score.

In conclusion, neither method is perfect alone. The best approach is a blend, where AI handles routine and objective checking quickly, and teachers review complex work and guide students. This combination keeps assessment fast, fair and meaningful.

Question 30 (b) Personal email: Orientation programme

Selected Question: 30.b

To: [friend's email address]

Subject: My first-year orientation programme at college

Dear [Friend's name],

How are you? I hope you are doing well and enjoying your college life too. I have been wanting to write to you about my orientation programme, so here it is.

The programme was held for the first-year engineering students at our college. It began with a welcome address by the Principal, followed by talks from senior faculty and some alumni. We also had fun icebreaker games and team activities, which helped me make new friends within the first day itself.

The best part was the tour of the academic facilities. We visited the library, the computer labs, the workshops and the seminar halls, and the staff explained how to use each of them. We were also told about the college rules on attendance, discipline, dress code and examinations, so now I know exactly what is expected from us.

Overall, the programme made me feel much more comfortable in the new environment. I understood how college life works and how to plan my studies, and I felt ready to start my academic journey with confidence.

Write back soon and tell me about your college too. Take care!

Your friend,
Harshini Thiyagarajan Logeshwari

Worksheet 2 (Model – I Practice Worksheet) – Exercises 1 to 15

Exercise 1: Simple Present Tense (Subject-Verb Concord)

1. recognize
2. requires
3. transmit
4. agrees
5. form

Exercise 2: Simple Past Tense

1. conducted
2. deployed
3. opened
4. published
5. detected

Exercise 3: Comparative Adjectives

1. higher
2. safer
3. faster
4. more durable
5. lower

Exercise 4: Modals of Obligation and Necessity

1. must
2. must

3. have to
4. must
5. has to

Exercise 5: Modals of Capability and Possibility

1. can
2. may
3. can
4. can
5. can

Exercise 6: Relative Clauses

1. The civil engineer who designed the suspension cable bridge received the national infrastructure award.
2. The spectrometer, which analyzes infrared emission spectra, is mounted inside the cleanroom vacuumchamber.
3. The technical specialist whose patent on pneumatic grippers was approved was reassigned to the roboticslab.
4. This microcontroller, which features ultra-low power sleep states, is ideal for battery-operated remotesensors.
5. The field researchers who surveyed the wind turbine farm submitted their acoustic emission assessmentyesterday.

Exercise 7: If-Clauses (Type 1)

1. exceeds
2. fails
3. engages
4. rise
5. registers

Exercise 8: If-Clauses (Type 3)

1. had started
2. had lubricated
3. would have been prevented
4. would have tripped
5. had remained

Exercise 9: Infinitives of Purpose

1. The mechanics installed dual damper springs to isolate chassis resonance.
2. Technicians purge the fuel line with dry nitrogen gas to remove residual flammable vapors.
3. We introduced an optical encoder to determine spindle velocity accurately.
4. The developers wrote an automated load testing script to simulate ten thousand concurrent connections.
5. Gold plating is applied to terminal connectors to prevent atmospheric corrosion.

Exercise 10: Gerunds after Prepositions

1. redesigning
2. replacing
3. reducing
4. cleaning
5. filtering

Exercise 11: Impersonal Passive Voice

1. 50 mL of hydrochloric acid was poured into the titration flask.
2. The robotic arm was tested under a 50 kg dynamic payload.
3. The optical pyrometer was calibrated prior to ignition.
4. Severe cavitation damage was observed along the impeller blades.
5. Volatile solvents are stored in flame-resistant containment cabinets.

Exercise 12: Adverbs of Manner

1. precisely
2. periodically
3. rapidly
4. firmly
5. smoothly

Exercise 13: Reported Speech

1. The quality manager asked the technician whether the insulation resistance test had been documented.
2. The lead architect enquired whether the server cluster could handle double the transaction throughput.
3. The field auditor asked the safety engineer whether he had inspected the gas leakage relief lines.
4. The client asked the team whether the firmware update would fix the intermittent battery drain.
5. The instructor asked the lab group why the multimeter had shown fluctuating voltage readings.

Exercise 14: Countable and Uncountable Nouns

1. Incorrect: "machineries" and "equipments". Correct: The procurement officer purchased several new piecesof machinery and equipment for the workshop.
2. Incorrect: "datas". Correct: The laboratory generated a huge amount of test data during the wind tunnel trials.
3. Incorrect: "softwares". Correct: We need three more software licenses to complete the finite element analysison the model.
4. Incorrect: "informations". Correct: The field report provided a lot of valuable information regarding the soilcomposition.
5. Incorrect: "apparatuses" and "glasswares". Correct: All chemical apparatus and glassware must be washedthoroughly after the experiment.

Exercise 15: Comparative Adjectives (Irregular Forms)

1. worse

2. better
3. farther
4. less
5. further

Worksheet 2 – Part B (Exercises 1 to 10)

Exercise 1: Simple Past Tense Conversion

1. The technician inspected the solar arrays and recorded the peak power conversion value.
2. The chemical analyzer detected trace mercury levels and sent an automated alert to the environmentaldashboard.
3. Maintenance crews lubricated the heavy conveyor bearings and replaced the worn rubber belts before themorning shift.
4. The supervisory software polled field sensors every five milliseconds and archived temperature trends into asecure database.
5. The testing team applied a 500-volt dielectric pulse across the stator winding and monitored for insulationbreakdown.

Exercise 2: Comparative Sentences

1. Cloud storage has a lower initial deployment cost than on-premise physical servers because no hardware hasto be bought. It is also more elastic, since its capacity can be increased within minutes.
2. Fiber-optic cables offer a much higher data transfer bandwidth than copper twisted-pair cables. They are alsoless susceptible to electromagnetic interference than copper cables.
3. Electric vehicles produce lower tailpipe emissions than internal combustion engine vehicles, as they releasenone at all. However, EVs generally have a higher initial purchase price than ICE vehicles.
4. Hydraulic actuators produce a greater raw force output than electromechanical actuators. They also carry ahigher risk of fluid leakage, because electromechanical actuators use no fluid.
5. Relational (SQL) databases have a more rigid schema than document (NoSQL) databases. Documentdatabases usually provide higher horizontal query throughput than relational databases.

Exercise 3: Relative Clauses

1. The algorithm that analyzes microscopic cracks in turbine blades was patented by the aeronautics team.
2. The lead chemical engineer, who noticed an abnormal pressure spike in chamber two, authorized theemergency boiler shutdown.
3. The laboratory houses a 400 kV impulse voltage generator, where high-voltage insulator puncture tests areconducted.
4. We engaged an external cybersecurity consultant whose firmware vulnerability audit exposed three backdoorports.
5. The supervisory system, which utilizes an artificial neural network, predicts transformer insulation life basedon dissolved gas concentrations.

Exercise 4: If-Clauses

1. (a) If the operating temperature exceeds 85°C, the cooling system will switch on automatically.
   (b) If the operator had pressed the emergency shutdown switch immediately, the accident would have beenavoided.
2. (a) If the supply voltage drops below the 200V threshold, the backup power supply will take over.
   (b) If the network administrator had enabled multi-factor authentication, the unauthorized access would

havebeen prevented.

3. (a) If toxic carbon monoxide concentrations surpass 50 ppm in the tunnel, the ventilation fans will start and analarm will sound.
   (b) If the mechanical team had checked the bearing lubrication schedule, the bearing would not haveoverheated.
4. (a) If the autonomous guided vehicle (AGV) detects an obstacle within 1 meter, it will stop immediately.
   (b) If the thermal sensor had registered the runaway reaction in time, the explosion would have beenprevented.
5. (a) If the software receives an unauthenticated API token, it will reject the request and return an errormessage.
   (b) If the civil inspectors had reviewed the core sample density tests, they would have detected the weakfoundation before construction.

Exercise 5: Impersonal Passive Voice

1. The crude extract was distilled at a constant temperature of 60°C.
2. The cross-section of the weld seam was polished with fine diamond paste.
3. Fifteen cast-iron pump casings were rejected due to internal porosities.
4. The resistance across the bridge circuit was measured using a calibrated digital ohmmeter.
5. The emergency aerodynamic braking flaps were initiated at an altitude of 3,000 meters.

Exercise 6: Reported Speech

1. The safety supervisor asked the intern whether he had verified the grounding connections on the generator.
2. The plant director asked the site engineer whether the emergency effluent treatment plant had receivedclearance.
3. The IT security lead enquired of the developer why he had disabled the firewall rule on the database server.
4. The client asked the project manager whether the bridge structure could withstand Category 4 hurricanewinds.
5. The research guide asked the postgraduate scholar how many sample trials he had conducted undercryogenic conditions.

Exercise 7: Countable vs. Uncountable Nouns

1. (a) We need to install several new pieces of software and machinery in the CAD laboratory.
   (b) The team collected plenty of useful information from the field test.
2. (a) The technician ordered ten new pieces of apparatus and modern electronic equipment for the antennalab.
   (b) High-speed train operations require advanced transport infrastructure and dedicated electrical power.
3. (a) The environmental sensor gathers extensive atmospheric data and pollution indices every minute.
   (b) The construction crew unloaded tons of sand, cement, and steel debris at the worksite.
4. (a) All electrical hardware and wiring must comply with local safety legislation.
   (b) The data scientist discovered several interesting insights after running the machine learning clustering.
5. (a) Laboratory staff must clear away all glassware and acidic waste before leaving the premises.
   (b) The project failed due to a lack of proper funding and technical research.

Exercise 8: Infinitives of Purpose and Gerund Constructions

1. (a) A pressure relief valve is installed to vent excess boiler steam.
   (b) The automated script runs every midnight to archive legacy server logs.
2. (a) Automotive engineers installed an intercooler to reduce the compressed air intake temperature.

(b) The technicians analyzed the structural stress points by placing piezoelectric strain gauges along thegirder.

3. (a) High-performance heatsinks feature copper heatpipes to conduct heat away from the silicon die.
   (b) The factory cut water consumption by 30% by implementing a closed-loop recycling filtration system.
4. (a) Safety guards are mounted over rotating flywheel belts to prevent accidental worker contact.
   (b) The algorithm optimizes memory bandwidth by pre-fetching sequential array indices into L2 cache.
5. (a) The electronics lab installed an uninterruptible power supply (UPS) to protect servers against suddenvoltage sags.
   (b) The metallurgy group enhanced surface hardness by diffusing nitrogen atoms into the steel casing.

Exercise 9: Adverbs of Manner

1. (a) automatically(b) flawlessly
2. (a) accurately

(b) instantly

3. (a) continuously
   (b) cautiously
4. (a) consistently
   (b) catastrophically
5. (a) periodically
   (b) smoothly

Exercise 10: Modals

1. (a) can
   (b) must not
2. (a) must
   (b) may
3. (a) must not
   (b) could
4. (a) should
   (b) must
5. (a) can
   (b) must not

Worksheet 2 – Topic 1: Writing Imperative Safety Instructions

Question 1: Chemistry and Materials Testing Laboratory

1. Wear a lab coat, safety goggles and nitrile gloves at all times inside the laboratory.
2. Handle concentrated acids and alkalis only inside the fume hood.
3. Add acid to water slowly; never add water to acid.
4. Do not smell, taste or touch any chemical directly.
5. Label every container clearly, and never return unused chemicals to the stock bottle.
6. Clean all glassware and testing equipment after each experiment to avoid contamination.

7. Keep food, drinks and open flames away from the work area.
8. Report every spill, burn or breakage to the instructor at once, and use the eyewash station or safety shower inan emergency.

Question 2: High-Voltage Electrical Machines and Drives Laboratory

1. Make sure the main supply is switched off before making or changing any connection.
2. Wear insulated rubber gloves and safety shoes, and stand on an insulated rubber mat near live panels.
3. Never touch exposed terminals, bare wires or rotating parts while the machine is running.
4. Get the circuit connections checked by the instructor before switching on the supply.
5. Know the location of the circuit breakers, fuses and emergency stop button, and keep them in workingcondition.
6. Remove rings, watches and metal jewellery, and avoid loose clothing, to prevent arc flash burns andaccidental short circuits.
7. Keep water and wet hands away from electrical equipment, and use only insulated tools.
8. Increase the motor load gradually, and switch off the supply and discharge the capacitors beforedisconnecting the circuit.

Question 3: Advanced Manufacturing and CNC Machining Workshop

1. Wear safety glasses, safety shoes and close-fitting clothes, and tie back long hair.
2. Keep the machine door and guard closed while the spindle is running.
3. Do not touch the spindle, cutter or workpiece until the machine has stopped completely.
4. Clamp the workpiece and tool holder firmly before starting the program.
5. Use a brush or chip hook, and heat-resistant gloves, to remove hot metal chips, and only after stopping themachine.
6. Wipe up coolant spills immediately, and keep the floor dry to prevent slipping.
7. Run the program in simulation mode first, and keep your hand near the emergency stop button during the firstcut.
8. Never leave the machine unattended or operate it without the instructor's permission.

Worksheet 2 – Topic 2: Free Writing – Festivals and Cultural Celebrations

Question 1: A traditional harvest festival and eco-friendly practices

Pongal: The Harvest Festival of Tamil Nadu and How Eco-Friendly Practices Can Protect Its Spirit

Pongal is the most important harvest festival of Tamil Nadu. It is celebrated in mid-January to thank the Sun God, the rain and the cattle for a good harvest. The festival lasts four days: Bhogi, Thai Pongal, Mattu Pongal and Kaanum Pongal.

On Bhogi, families clean their homes and throw away old things. On Thai Pongal, they draw colourful kolams at the entrance and cook fresh rice and milk in a new clay pot in the open yard. When the milk boils over, everyone shouts "Pongalo Pongal!" The pot is decorated with turmeric plants and sugarcane. The festive meal includes sakkarai pongal, venn pongal, sambar, vadai and payasam, served on banana leaves. On Mattu Pongal, farmers decorate and worship their cattle, and Kaanum Pongal is a day for family visits and community games.

Modern eco-friendly practices can preserve the core spirit of the festival. Instead of burning plastic and old tyres on Bhogi, communities can burn only dry leaves or turn them into compost, which reduces smoke and air pollution. Clay pots, banana leaves and areca-leaf plates can replace plastic and thermocol, and kolams can be drawn with natural rice flour. Villages can organise tree-planting drives, rainwater harvesting and cloth-bag campaigns during the festival, and use solar lamps instead of excess electric decorations.

In this way, Pongal can remain joyful and traditional while staying kind to nature, and the next generation will learn that celebration and conservation can go together.

Question 2: A temple car festival and sustainable measures

The Temple Car Festival: Faith, Food and a Cleaner Celebration

In my hometown, as in many towns of Tamil Nadu, the temple car festival, or Ther Thiruvizha, is the biggest event of the year. Thousands of devotees gather in the streets to watch the deity pulled in a huge, richly decorated wooden chariot, called the ther.

The festival begins with flag hoisting and several days of processions. On the car day, devotees of all ages hold thick ropes and pull the heavy chariot together through the four main streets, chanting prayers while drums and nadaswaram fill the air. This shared effort is a strong symbol of unity, because people from every community take part. Women draw kolams on the road, and families offer coconuts, bananas and flowers to the deity. Volunteers give free food such as pongal, puliyodharai, curd rice and sweet payasam as prasadam, served on leaves.

However, such a large crowd also creates problems. Single-use plastic cups, plates and bags litter the streets, and loudspeakers cause noise pollution. Eco-friendly measures can solve these problems. Organisers can ban single-use plastic and provide steel tumblers, banana leaves and paper bags. Dustbins and waste-sorting points can be placed along the route, and volunteers can collect and compost the food waste. Noise can be controlled by limiting loudspeaker volume and using traditional instruments. Barricades, marked entry and exit lanes, volunteers, and first-aid points can manage the crowd safely.

With these steps, the festival can keep its sacred joy and remain clean and peaceful.

Question 3: Regional folk art and handicraft fairs with renewable energy and zero-waste practices

Folk Art and Handicraft Fairs: Heritage with a Green Future

A regional folk art and handicraft fair is a colourful meeting of tradition and community. Walking through the fair grounds, visitors see Karagattam and Oyilattam performances, hear the beat of the parai drum, and watch artisans make Thanjavur dolls, palm-leaf baskets, clay pots and silk sarees. The air smells of fresh filter coffee, murukku, adhirasam and hot sundal.

These fairs are important because they give artisans a market and help young people see the value of their heritage. Many stalls offer live demonstrations, so children can try pottery or weaving themselves. Local delicacies prepared by village cooks add to the festive atmosphere and support small businesses.

However, large fairs use a lot of electricity and produce a lot of waste. Modern environmental strategies can solve this. Solar micro-grids, made of rooftop panels and batteries, can power the lights, sound systems and stalls without diesel generators. Zero-waste composting can turn leftover food and leaves into manure for local farms. Organisers can use reusable steel plates, clay cups and cloth banners, provide separate bins for wet and dry waste, and recycle paper and cardboard.

If we combine the traditional skill of our artisans with clean energy and responsible waste management, these fairs can protect our culture and our planet together.

Worksheet 2 – Topic 3: Free Writing – Food and Technical Travel Plans

Question 1: Technical study tour to a major hydroelectric dam

A Technical Study Tour to the Mettur Hydroelectric Dam

Our department has planned a one-day technical study tour to the Mettur Dam on the Cauvery River in Salem district. The group of forty students and two faculty members will leave the college by a chartered bus at 4:30 a.m. and travel along the national highway, with a short breakfast halt on the way. The journey takes about six hours, and we will return by night. Students will carry ID cards, water bottles and a notebook.

After registering at the gate, an engineer will take us through the powerhouse. The long concrete dam holds back the huge Stanley Reservoir. Inside the powerhouse, we will see the penstocks, the turbines and the generators, and learn how falling water turns turbines to produce electricity. We will also observe the control room, where operators monitor

water level, flow and power output on screens, and the spillway gates used to release extra water safely.

For lunch, we will try local dishes such as hot rice with sambar, fish fry from the river and fresh mango juice, which make the trip a cultural experience as well. In the evening, we will walk through the nearby park and enjoy the green landscape and the calm reservoir.

The visit will help us understand renewable energy, water management and the role of civil, mechanical and electrical engineering in one project. We will follow all safety rules, wear helmets where needed, and stay with our group at all times.

Question 2: Technical study tour to an integrated solar and wind hybrid energy farm

A Technical Study Tour to a Solar and Wind Hybrid Energy Farm

We plan a two-day study tour to the Muppandal wind farm region near Kanyakumari, where wind turbines and solar panels work together. Thirty students and two faculty members will take the overnight train from Chennai to Nagercoil, then travel about one hour by a hired van to the farm. We will stay one night in a budget lodge nearby, and return by train on the second night.

On arrival, a site engineer will give a short briefing on how the hybrid plant works. The wind turbines turn on tall towers, and their blades rotate a generator inside the nacelle. We will watch how the strong coastal wind drives them, and how the blades adjust their angle with the wind speed. In the solar section, we will observe rows of photovoltaic panels, inverters and transformers that feed power into the grid. The control room will show how the two sources balance each other: solar power peaks in the afternoon, whereas wind is strong in the evening and

night.

For food, we will taste Nanjil Nadu specialities such as fish curry with rice, banana chips, avial and fresh coconut sweets. Eating in small local hotels along the route will also help us understand the region's culture.

The tour will show us the practical side of renewable energy, and we will follow safety rules by wearing helmets, keeping away from the turbine bases and obeying the engineers' instructions.

Question 3: Technical field visit to an automated seawater reverse osmosis desalination plant

A Technical Field Visit to a Seawater Reverse Osmosis Desalination Plant

Our class is planning a half-day field visit to a seawater reverse osmosis desalination plant on the East Coast Road near Chennai, such as the Nemmeli plant. About fifty students and three faculty members will travel in two college buses, leaving at 8:00 a.m. and reaching the plant in about ninety minutes. We will carry ID cards and wear closed shoes for safety.

At the plant, the engineers will first explain the process. We will see the intake pumping station, which draws seawater through long pipes under the sea bed. The water then passes through pre-treatment filters that remove sand and suspended particles. Next, high-pressure pumps push the water through the membrane filtration racks, where the salt is separated and fresh water is produced. We will also see the brine discharge outlet, where the concentrated salt water is carefully diluted and released back into the sea to protect marine life. The control room shows the whole process through automated screens.

After the visit, we will have lunch at a local seaside restaurant. The coastal menu includes fish fry, prawn masala, crab curry with steamed rice and filter coffee, which gives us a taste of the Chennai coastal culture.

The visit will help us see how engineering solves water scarcity, how energy is used in the process, and why environmental care in brine disposal is important.

Worksheet 2 – Topic 4: Formal Letters of Request

Question 1: Letter to the Head of the Computer Science Department (lab access for a hackathon)

Harshini Thiyagarajan Logeshwari
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Head of the Department
Department of Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for permission and lab access after college hours to prepare for a National AI Hackathon

Respected Sir/Madam,

I am writing on behalf of my project team of four first-year students. We have registered for a National AI Hackathon to be held on [date], and we respectfully request permission to use the computer laboratory after college hours.

The objective of our project is to build a simple AI-based solution that solves a real campus problem, using Python and machine learning. We need about three weeks to design, train and test the model, and our home systems are not powerful enough for this.

We require access to the lab between 4:30 p.m. and 7:30 p.m. on weekdays, with Python, Jupyter Notebook, TensorFlow or scikit-learn, Git and a stable internet connection. We promise to follow all lab rules, maintain the equipment carefully, and leave the lab clean and switched off after use. We will also work under the supervision of a faculty member if required.

We would be very grateful if you could grant us this permission.

Thank you for your kind consideration.

Yours obediently,
Harshini Thiyagarajan Logeshwari

(On behalf of the project team)

Question 2: Letter to the Plant Manager (industrial plant visit)

Harshini Thiyagarajan Logeshwari
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Plant Manager
[Name of Factory], Industrial Robotics and Automation Division [Address]

Subject: Request for permission to conduct an industrial visit for 50 first-year engineering students Respected

Sir/Madam,

I am a first-year engineering student of St. Joseph's College of Engineering, Chennai. On behalf of my class, I request your permission to visit your robotics and automation factory with fifty students and two faculty members on a convenient date in [month].

The intended learning outcomes of the visit are to understand how robots and automated systems work on a real production line, to see how sensors, controllers and robotic arms are combined, and to learn about industrial safety practices and quality control. The visit will also help us connect our classroom learning with practical engineering and

inspire us to think about careers in automation.

We will arrive in a group, carry our ID cards, follow all safety rules, and obey your staff's instructions throughout. We would be glad if a short briefing and a guided tour of about two hours could be arranged.

I would be grateful if you could kindly grant us this opportunity. I look forward to your reply.

Yours faithfully,
Harshini Thiyagarajan Logeshwari

Question 3: Letter to the Head of the Mechanical Engineering Department (machine shop access for an electric go-kart)

Harshini Thiyagarajan Logeshwari
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Head of the Department
Department of Mechanical Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for permission and machine shop access after college hours to build an electric go-kart

Respected Sir/Madam,

Our student team has been selected to take part in an inter-collegiate design challenge to be held on [date], where we must build and test an energy-efficient electric go-kart. We therefore request permission to use the machine shop after regular college hours.

We need the workshop to cut and weld the chassis, fit the motor, battery and braking system, and test the kart on the campus ground. We estimate that we will need about four weeks, working from 4:30 p.m. to 7:30 p.m. on weekdays and during part of the weekends.

We will work only under the supervision of the workshop instructor, wear safety glasses, gloves and shoes at all times, and follow all machine safety rules. We will take care of the tools and keep the shop clean after use.

We would be thankful if you could grant us this permission and allow us to bring honour to the college.

Thank you for your kind support.

Yours obediently,
Harshini Thiyagarajan Logeshwari

(On behalf of the team)

Question 4: Letter to the Principal / Dean of Student Affairs (sponsorship and duty leave)

Harshini Thiyagarajan Logeshwari
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Principal / Dean of Student Affairs
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for financial sponsorship and duty leave to present a drone surveillance prototype at the National Defense Innovation Expo

Respected Sir/Madam,

I am happy to inform you that a five-member multidisciplinary team of our college has been selected to present a drone surveillance prototype at the National Defense Innovation Expo, to be held on [dates] at [venue]. We respectfully request financial sponsorship and duty leave for this event.

Our prototype uses a camera-equipped drone with image processing to monitor sensitive areas in real time. Presenting it at a national level will give us valuable feedback from experts and will bring recognition to our college.

We request sponsorship for travel, accommodation, registration and prototype transport, which we estimate at ₹ [amount], and duty leave for [number] days so that we do not lose attendance. We will make up for the missed classes and submit a report after the event.

I would be grateful for your kind support and approval.

Thank you.

Yours obediently,
Harshini Thiyagarajan Logeshwari
(On behalf of the team)

Worksheet 2 – Topic 5: Compare and Contrast Descriptive Essays

Question 1: Electric Vehicles versus Internal Combustion Engine Vehicles

Electric Vehicles versus Conventional Internal Combustion Engine Vehicles

Electric vehicles (EVs) and internal combustion engine (ICE) vehicles are two different answers to the same need, which is transport. Each has clear positives and negatives.

Consider battery life first. An EV battery usually lasts eight to ten years, but replacing it is very expensive. In contrast, an ICE vehicle has no battery problem, though its engine needs regular servicing. As for the manufacturing footprint, EVs create more emissions during production because of battery mining and processing, whereas ICE vehicles have a lighter manufacturing footprint. However, over its lifetime an EV

produces far lower emissions, especially when the electricity comes from renewable sources.

Operating costs also differ. EVs have fewer moving parts, so they need less maintenance, and electricity per kilometre is cheaper than petrol or diesel. On the other hand, ICE vehicles cost less to buy, and their repair shops are available everywhere. Fuel prices, however, keep rising and make them costly to run.

Charging network feasibility is the biggest challenge for EVs. Charging stations are still limited, especially on highways and in small towns, and charging takes much longer than refuelling. ICE vehicles, conversely, can be refuelled in minutes at petrol stations spread across the country.

In conclusion, EVs are cleaner and cheaper to run, whereas ICE vehicles are cheaper to buy and more convenient at present. As battery prices fall and charging networks grow, EVs are likely to become the better choice for the future.

Question 2: Remote/Virtual Internships versus On-Site Internships

Remote Internships versus On-Site Internships for First-Year Engineering Students

Internships help early-stage engineering students connect classroom theory with real work. Today, students can choose between remote (virtual) and on-site (physical) internships, and each has advantages and disadvantages.

Hands-on skill acquisition is the biggest difference. On-site internships give direct access to machines, laboratories and real equipment, so students learn practical skills such as wiring, fabrication and testing. Remote internships, in contrast, mostly involve software, simulations and online tasks, which are useful for coding and design but cannot replace physical practice.

In terms of schedule flexibility, remote internships win clearly. Students can work from home, save travel time and cost, and manage studies alongside the internship. On-site internships, on the other hand, demand fixed hours and daily

travel, which can be tiring during the semester.

Mentorship is stronger on-site, because supervisors can guide students directly and correct mistakes immediately. Online mentors, however, are often busy and communicate only through scheduled calls or messages, so doubts may stay unresolved for days.

Professional networking also favours on-site internships, as students meet colleagues, build relationships and learn workplace culture naturally. Remote interns may never meet their team face to face, although online platforms still allow them to connect with professionals from different cities.

In conclusion, remote internships are better for flexibility and software skills, whereas on-site internships are better for practical learning, mentorship and networking. A first-year student should ideally begin with a remote internship and move to an on-site one later.

Question 3: Cloud Computing Infrastructure versus On-Premise Data Centers

Cloud Computing versus On-Premise Physical Data Centers for Engineering Enterprises

Engineering enterprises must decide whether to run their systems on the cloud or in their own physical data centers. Both choices have strengths and weaknesses.

Upfront capital expenditure is the clearest difference. Cloud services need almost no initial investment, because companies simply pay for what they use. On-premise data centers, in contrast, require heavy spending on servers, cooling systems, power backup and buildings before any work begins.

As for recurring overheads, cloud bills can grow quickly if usage is not monitored, whereas an on-premise setup has predictable costs once it is built, although it still needs staff, electricity and maintenance. Therefore, cloud suits changing workloads, while on-premise suits stable and long-term demand.

Workload scalability strongly favours the cloud. Companies can add computing power in minutes during simulations or product launches and reduce it afterwards. On the other hand, on-premise systems have fixed capacity, so expanding them takes weeks or months and leaves hardware idle during quiet periods.

Data security and compliance are more complex. On-premise data centers give the company full physical control, which is important for defence, healthcare or confidential design data. Cloud providers offer strong security tools and certifications, but the company depends on a third party and must follow data-location laws

carefully.

In conclusion, the cloud is better for flexibility, speed and low starting cost, while on-premise infrastructure is better for control and predictable long-term expenses. Many enterprises now choose a hybrid model that combines the advantages of both.

Question 4: 3D Printing versus Traditional CNC Milling

3D Printing (Additive Manufacturing) versus CNC Milling (Subtractive Manufacturing) in Prototype Fabrication

In prototype fabrication, 3D printing and CNC milling follow opposite principles. A 3D printer builds a part layer by layer, whereas a CNC machine cuts material away from a solid block. Each method has positives and negatives.

Material wastage is the first point of contrast. 3D printing uses only the material needed for the part, so wastage is low. CNC milling, on the other hand, removes a large amount of material as chips, which increases the cost of expensive metals.

When it comes to production speed, the picture changes with batch size. For single prototypes and small batches, 3D printing is faster because it needs no tooling or setup. For larger batch runs, however, CNC milling is quicker, as it can produce identical parts continuously at high speed.

Surface finish and accuracy favour CNC milling. It produces smooth surfaces and very tight tolerances, so parts often

need no extra work. In contrast, 3D printed parts often show layer lines and need sanding or polishing, and their accuracy is lower.

Geometric complexity is the strength of 3D printing. It can easily create hollow parts, internal channels and lattice structures that are impossible to cut with a rotating tool. CNC milling is limited by tool access and the direction of cutting.

In conclusion, 3D printing is ideal for complex, low-volume prototypes with little waste, whereas CNC milling is better for accurate, smooth and larger-batch parts. Engineers often use both methods together to get the best result.

Worksheet 2 – Topic 6: Personal Email Writing

Question 1: Email to a high school mentor/teacher about the first semester

To: [teacher's email address]

Subject: My first semester at engineering college

Dear Sir/Madam,

I hope you are doing well. I wanted to write to you because your guidance in school gave me the confidence to start my engineering journey, and I would like to share how my first semester has been.

The biggest change was the move to independent campus life. Nobody reminds us about deadlines now, so I had to learn to manage my own time, plan my study hours and take responsibility for my work. It was difficult at first, but I am getting used to it.

In our laboratory sessions, we worked on our first small projects, such as basic programs in Python and C and simple circuits. Seeing my own code run correctly gave me a great sense of satisfaction, and I realised how different practical learning is from reading theory.

For the end-semester examinations, I have made a timetable that gives equal time to every subject. I revise the notes every weekend, solve previous question papers and practise the lab experiments so that I do not forget the procedures.

Thank you for everything you taught me. I will be very happy to hear from you and to receive any advice for the coming semesters.

With respect and regards,
Harshini Thiyagarajan Logeshwari

Question 2: Email to a high school science teacher about the first term

To: [science teacher's email address]

Subject: Reflections on my first term in engineering

Dear Sir/Madam,

I hope you are keeping well. It has been a few months since I joined engineering college, and I wanted to share my first-term experience with you, since you were the one who made science exciting for me.

The first thing I noticed was how different the hands-on lab coursework is. In school, we mostly saw experiments in a few periods, but here we handle components, tools and software ourselves. I made mistakes in the beginning, but each mistake taught me something and I am becoming more careful and confident.

I also took part in my first technical club activity and a small hackathon. Working with a team to build a solution within a limited time was challenging and thrilling, and I learned a lot about teamwork and debugging from my seniors.

For the upcoming university theory and practical exams, my strategy is simple. I will finish revising each unit at least two weeks before the exams, practise numerical problems daily, and repeat the lab experiments so that I can answer viva questions without fear.

I would be grateful for any advice you may have. Please take care.

Yours sincerely Harshini Thiyagarajan Logeshwari

Question 3: Email to an elder sibling/alumnus seeking advice

To: [sibling's or alumnus's email address]

Subject: Need your advice on balancing studies and technical clubs

Dear [Name],

How are you? I hope college and work are going well. I am writing because I need your advice on something I am struggling with.

My day starts at 6:30 a.m., and I attend classes until 4:30 p.m. After that, I go to my technical club meetings for about two hours, and in the evening I try to study and finish my assignments. By the time I am free, I am usually very tired.

My biggest challenge is managing the lab record submissions. Because the records must be completed every week, they pile up, and I often finish them at the last minute, which affects the quality.

Since you have handled the same situation, could you please tell me how you balanced your academics with the clubs? I would also like your tips for semester exam revision, such as how to plan the timetable, which subjects to start with, and how to use previous question papers.

I am looking forward to your reply. Thank you in advance, and please take care.

Your loving brother/sister,

Harshini Thiyagarajan Logeshwari

Worksheet 2 – Topic 7: Travel Planning and Budgetary Itineraries

Question 1: 4-day tour to an aerospace and satellite launch station

A 4-Day Student Technical Tour to an Aerospace and Satellite Launch Station

Our department plans a four-day tour to a satellite launch station for forty students and two faculty members. We will travel by a chartered train coach and local bus, so that the cost is reasonable.

Budget (per student): train fare ₹1,200, local bus ₹400, accommodation for three nights ₹1,500, food ₹1,200, entry and miscellaneous charges ₹400. The total is ₹4,700, which gives about ₹1,88,000 for the full group of forty.

Day 1: Overnight travel and arrival, check-in at the guest house and an orientation briefing. **Day 2:** Visit the launch complex viewing gallery, the rocket assembly area and the mission control centre, where scientists explain how a launch is planned and tracked. **Day 3:** Visit the satellite integration and testing laboratory and the space museum, followed by an interaction with engineers on careers in aerospace. **Day 4:** Morning group discussion, preparation of our tour report, and the return journey.

Safety protocols: Carry your ID cards and keep them visible at all times. Follow the instructions of the security staff, avoid photography in restricted areas, and stay with the group. Wear closed shoes and drink plenty of water. Keep the emergency contact numbers of the faculty coordinators and follow the launch-site rules strictly.

This tour will give us a first-hand view of how rockets and satellites are built and operated.

Question 2: 4-day tour to a major commercial port and maritime dry-dock facility

A 4-Day Industrial Exposure Tour to a Major Commercial Port and Dry-Dock Facility

Our class of forty first-year students and two faculty members will visit a major commercial port and a dry-dock facility over four days. We will travel in a hired bus, and stay in a budget hotel near the port.

Estimated cost per student: travel ₹1,500, stay for three nights ₹1,800, food ₹1,200, entry passes and miscellaneous ₹500. The total is ₹5,000 per student.

Day 1: Travel to the port city, check in and a short orientation. **Day 2:** Technical briefing at the port authority, followed by a visit to the container terminal, where cranes, conveyors and tracking systems handle cargo. **Day 3:** Visit the dry-dock, where ships are lifted and repaired, and see how hull cleaning, welding and engine maintenance are done. An engineer will also brief us on ship stability. **Day 4:** Visit the control tower and the vessel traffic system, a feedback session and the return journey.

Dockyard safety protocols: Wear helmets, safety shoes and high-visibility jackets at all times. Do not go near moving cranes, cargo or the edge of the dock. Stay in the group, obey the guides and never touch machinery. Keep emergency contacts ready, and carry first-aid supplies.

This tour will help us understand how ports and shipyards work in real life.

Question 3: 4-day tour to an electric locomotive manufacturing works and railway signalling center

A 4-Day Technical Study Tour to an Electric Locomotive Works and a Railway Signalling Center

We plan a four-day study tour for thirty-five students and two faculty members to an electric locomotive manufacturing works and an automated railway signalling center. We will book sleeper-class seats on trains for both journeys, and use a local bus for the visits.

Category-wise expenditure per student: rail tickets ₹1,400, local transport ₹300, accommodation for three nights ₹1,500, food ₹1,200, and entry and miscellaneous ₹300. The total is ₹4,700.

Day 1: Overnight train journey and arrival. **Day 2:** Visit the locomotive works, where we will see the shop-floor stages such as shell fabrication, wiring, traction motor assembly and final testing of the engine. **Day 3:** Visit the automated signalling center, where engineers explain how sensors, track circuits and computers control train movement and keep the trains safe. **Day 4:** Interaction with engineers, preparation of a tour report and the return journey.

Site safety rules: Wear helmets, safety shoes and closed clothing in the shop floor areas. Never walk on the railway tracks, and keep a safe distance from cranes and live electrical equipment. Follow the guides' directions and stay with the group. Do not touch any machine or signal equipment, and report any problem immediately to the faculty coordinator.

This tour will connect our classroom learning with real railway engineering.

Project Proposal: Ideation

LunArk: Genesis, Technical Specification

Team LunArk | NASA Space Apps Challenge 2026 | Draft v1, October 17, 2026

Status: draft. Items marked [default] are working assumptions the team can change. Items marked [open] still need a decision.

1. Overview

LunArk: Genesis is a desktop game where the player runs a lunar outpost. You place buildings, keep power flowing through the long lunar night, protect the base from radiation, earn money and stop the crew from getting bored. The aim is to let students aged 13-16 feel the engineering trade-offs that decide whether a real Moon mission succeeds or fails.

The game is built in Python with pygame-ce. All art is hand-drawn pixel art made in Aseprite. The code is written with help from an AI coding agent (Google Antigravity) and reviewed by the team, and the project submission discloses this.

2. Goals and non-goals

2.1 Goals

- Make real mission trade-offs tangible: power, life support, radiation shielding, money and crew morale.
- A 13-16 year old should understand it within about ten minutes, thanks to a guided tutorial.
- It runs on Windows, macOS and Linux with two commands and no installer.
- It is fully documented, so a judge or a new teammate can run, read and extend it.
- Game logic, user interface and game data stay strictly separate.

2.2 Non-goals

- No multiplayer, online features or accounts.
- No packaged executables (no PyInstaller). Players run the game from source.
- No AI-generated art, fonts or audio. The team makes all art.
- No zooming or rotating of sprites, which keeps the pixel art crisp.
- No real orbital mechanics. The science content is simplified, but it must be accurate.

3. Target audience and design feel

- Audience: students aged 13-16, with little or no engineering background.
- Feel [default]: calm and hopeful at the start, with pressure rising later, but never unfair. When players fail, they should think “I see what I did wrong”, not “that was random”. Failure screens always show the exact reason and a hint.
- Inspiration: the mobile game Spaceflight Simulator, used only for ideas. Those are a clean, flat, icon-first interface; building from parts with a ghost preview; a side-view 2D world; an always-visible time warp that slows itself down when something important happens; and predictive information such as countdowns and warnings. No art, code, name or exact UI is copied.

4. Technology stack

Item	Decision
Language	Python 3 (minimum tested version to be recorded in the README)
Library	pygame-ce, version pinned in requirements.txt
Testing	pytest (development only, in requirements-dev.txt)
Art tool	Aseprite, exported as PNG
Packaging	None. Run from source
Platforms	Windows, macOS, Linux, desktop only

4.1 How players run the game

```
pip install -r requirements.txt
python3 main.py
```

On Windows the second command may be python main.py or py main.py. The README documents all three, along with optional virtual-environment steps.

5. Repository structure

```
LunArkGenesis/    main.py                    entry point    requirements.txt
pygame-ce  only,  pinned      requirements-dev.txt      pytest      README.md
full documentation   LICENSE              GPL-3.0   fonts/              pixel
font (team-supplied)   img/                 all art (team-made)     logo/  ui/
icons/  terrain/  sky/  buildings/     characters/  cutscene/  events/  fx/
vehicles/    saves/                       runtime only, gitignored    tests/
pytest tests for game/logic   game/     app.py             window, main loop,
screen manager     constants.py          fixed numbers, colors, paths, index
tables      assets.py              image/font loading, placeholders, 9-slice
logic/                           game  rules,  no  pygame  imports         ui/
drawing and input     screens/             splash, title, settings, cutscene,
play, pause, end     data/                fixed game content as JSON
```

Rule of thumb: logic decides what happens, ui shows it, data holds the numbers and text, and saves holds the player's files.

6. Architecture

6.1 Layers

Layer	Responsibility	Rules
game/logic	Bars, clock, world grid, buildings, events, win/lose, settings, save/load	Never imports pygame. Fully unit-tested
game/ui	Screens, widgets, drawing, input, display scaling	Contains no game rules
game/data	buildings.json, events.json, balance.json, story.json	Numbers and text only
game/assets.py	Loads images and fonts	Draws a labelled placeholder when a file is missing
saves/	settings.json, savegame.json	Created at runtime, validated on load

6.2 Main loop

The app owns one window and a screen manager. Each screen implements handle events, update and draw. The simulation moves forward in fixed logic ticks (1 tick = 1 second at 1x speed), independent of the frame rate, so the FPS setting never changes game speed.

6.3 Error handling

A missing or corrupt save, settings file, JSON file or image must never crash the game. It falls back to defaults or placeholders and logs a clear message. At startup it prints a single summary of missing assets.

7. Display, fullscreen and settings

7.1 Rendering

- The game draws to an internal canvas of 480 x 270 [default] (16:9) with 16 x 16 pixel tiles.
- The canvas is scaled to the window or screen with nearest-neighbor scaling. The aspect ratio is kept, and black bars fill any gaps.
- Mouse positions are converted back to canvas coordinates.
- An optional integer scaling setting uses whole-number scale factors only, which gives sharper pixels but larger bars. Default: on.

7.2 Display modes

- The first launch starts in fullscreen at the desktop resolution, and the choice is remembered.
- F11 toggles between fullscreen and windowed mode from any screen.
- Esc never quits instantly. It goes back in menus and opens the pause menu in the game. Quit is a button on the title and pause menus.
- On Windows the process is made DPI-aware so sizes are correct on scaled displays.

7.3 Settings menu

Available from the title screen and the pause menu, by mouse or keyboard.

Setting	Options
Display mode	Fullscreen, Windowed

Setting	Options
Resolution (windowed size)	1280x720, 1366x768, 1600x900, 1920x1080, 2560x1440, Native desktop. Only sizes that fit the monitor are listed. Greyed out in fullscreen
FPS cap	30, 60, 120, 144, Unlimited. Default 60
VSync	On, Off. Default on. Hidden if unsupported
Show FPS counter	On, Off. Default off
Integer scaling	On, Off. Default on
Buttons	Apply, Back, Reset to defaults

Changing the display mode or resolution brings up a “Keep these settings? (reverting in 10s)” prompt that reverts on its own, so nobody gets stuck on a bad screen. Settings are saved to saves/settings.json, validated on load, and replaced by defaults if invalid.

8. Screen flow

1. Splash: a black screen. The logo (33 x 33 px, integer-scaled with nearest-neighbor) fades in and out over about 3 seconds. Any key or click skips it.
2. Title screen: wordmark and background. Menu: New Game, Continue (disabled without a save), Settings, Quit.
3. Intro cutscene: a full-screen picture and a dialogue box with typewriter text. The player wakes up to a man looking at them. He welcomes them to the Moon and asks their name (typed, maximum 12 characters, default “Commander”). He explains that NASA sent them to establish a base and introduces himself as Lieutenant Hale [default name], then walks through the first steps. The cutscene can be skipped.
4. Guided tutorial: inside the real game. The lieutenant waits for the player to finish each step: place a Fuel Cell Generator, place the Habitat Pod, watch the Energy bar, pan the camera, build a Lab, understand Boredom and build a Gym, understand the Event bar, read the sunset countdown, then dig a building underground and see its radiation exposure improve.
5. Main game.
6. Pause menu: Resume, Settings, Save, Quit to Title, Quit Game.
7. End screens: win or lose, always with the exact reason and a hint for next time. Buttons: Retry, Title.

9. Game design

9.1 The world

- A side-view cross-section of the Moon, 48 columns wide [default], scrolling horizontally on the 480 px canvas.
- From top to bottom: black sky (the lunar sky is black even in daytime), the ground line, then 4 rows [default] of underground regolith.
- The sky shows stars, the Sun crossing during the day, and Earth hanging almost still. Two parallax horizon layers add depth.

- The camera pans horizontally only (A and D, arrow keys, or dragging with the middle mouse button). There is no zoom.

9.2 Placing buildings

- The player picks a building from a collapsible build tray at the bottom of the screen.
- A ghost preview follows the mouse, snapped to the grid. It turns green when the spot is valid and red when it isn't, with a one-line reason such as "Needs the surface", "Not enough cash", "Locked: needs Science 50" or "Space is taken".
- Left click places, and right click or Esc cancels. Demolishing refunds part of the cost.
- Each building has a placement type: surface, underground or either.
- Building underground digs the footprint automatically. The cost is the building cost plus a dig cost per tile. No access shafts are modeled, which is a deliberate abstraction stated in the README.
- Radiation exposure runs from 1.0 on the surface down to about 0.05 when deep, depending on how many regolith rows sit above the building. A small radiation icon shows it.
- Selecting a building opens an info card with its role, energy in and out, exposure, status, a science fact, and Repair or Demolish buttons.

9.3 Time and day-night cycle

- 1 tick equals 1 second at 1x speed. Time warp buttons: Pause, 1x, 2x, 4x, 8x.
- Warp drops back to 1x automatically, with an on-screen reason, when a warning or event occurs or when Energy falls below 15% [default].
- The real lunar night lasts about 14 Earth days, so it is compressed: 180 ticks of sunlight, then 180 ticks of night [default], set in balance.json.
- The HUD shows a sun or moon icon and a countdown to the next sunset or sunrise.

9.4 The five bars

Bar	Meaning
Energy	Stored energy between 0 and a capacity. Each tick: production minus consumption. At 0, buildings shut down in priority order, with the Habitat Pod last
Science	Accumulated research points. Used for the win goal and for unlocking equipment tiers
Boredom	0 to 100. Rises over time with crew size. Activity buildings lower it
Revenue	Cash balance. Spent on building, digging and repairs, earned by Cashflow buildings. Shown as a bar toward the next unlock plus a number
Event	0 to 100. Drifts up when Boredom is high and down when Boredom is low. A higher value means a higher chance per tick of a random failure event

Bars flash when they run low or turn dangerous. All rates and thresholds live in balance.json, so they can be tuned without code changes.

9.5 Buildings

There are four categories with three tiers each. Tiers 2 and 3 unlock only when both Science and Revenue

pass thresholds set in balance.json.

Category	Tier 1	Tier 2	Tier 3	Rule
Generator	Fuel Cell Generator (2x2)	Solar Array (3x2)	Nuclear Plant (3x3)	Produces Energy
Cashflow	Datacentre (2x2)	Precious Metal Mining (3x2)	Helium-3 Extraction (3x3)	Uses Energy, produces cash

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HARSHINI THIYAGARAJAN LOGESHWARI

B.E. Computer Science and Engineering | Aspiring Cybersecurity Analyst

GITHUB	CAREER AMBITION	AREA OF INTEREST
https://github.com/tlharshini123-source	Cybersecurity Analyst	Networking & Information Security

ABOUT ME

Hi, I'm Harshini Thiyagarajan Logeshwari, a first-year B.E. Computer Science and Engineering student at St. Joseph's College of Engineering. I am interested in cybersecurity, especially computer networking and the ways digital systems can be protected. I enjoy exploring new technologies, learning how things work, and improving my programming and problem-solving skills.

I am currently building my foundation in C, Python and MySQL while learning the fundamentals of networking and information security. I believe that consistent practice, curiosity and hands-on projects will help me grow into a capable cybersecurity professional.

EDUCATION

Programme	B.E. Computer Science and Engineering
Institution	St. Joseph's College of Engineering, OMR, Chennai
Current Level	First Year | 2026–2030
School	Sai Ram Vidyalaya, Madipakkam

TECHNICAL SKILLS

Programming	C (beginner), Python (learning)
Database	MySQL – building foundational knowledge
Core Concepts	Computer networks, cybersecurity fundamentals, problem solving
Platforms & Tools	Git and GitHub; continuing to explore development tools

CAREER OBJECTIVE

My short-term objective is to strengthen my programming, networking and database fundamentals and gain practical experience through beginner-friendly projects and technical events. My long-term goal is to become a Cybersecurity Analyst and continue developing my knowledge in areas such as network security, threat awareness and secure computing.

Personal strengths: Adaptability • Leadership • Curiosity • Willingness to learn

PART B: GITHUB PROFILE README

A profile introduction ready to adapt for the GitHub README section.

Hi there, I'm Harshini Thiyagarajan Logeshwari ■

B.E. CSE Student | Cybersecurity Enthusiast | Interested in Networking

GitHub github.com/tlharshini123-source	**Goal** Aspiring Cybersecurity Analyst

ABOUT ME

- I'm a Computer Science and Engineering student building a strong foundation in programming and computing.
- I'm interested in cybersecurity and computer networking, and I want to understand how systems communicate andhow digital information can be protected.
- I'm currently learning C, Python and MySQL while improving my problem-solving and technical skills.
- I enjoy learning new concepts, exploring technology and taking part in activities that help me grow.

TECH STACK & TOOLING

Domain	Technologies / Focus
Programming	C, Python (learning)
Database	MySQL
Version Control	Git, GitHub
Cybersecurity	Security fundamentals, safe computing practices (learning)
Networking	Networking basics and communication concepts (learning)

FEATURED REPOSITORIES

GitHub Profile Repository

Use the profile README repository to introduce yourself, highlight your learning journey and organise links to projects as you publish them.

https://github.com/tlharshini123-source

Projects to showcase as they are completed

- C Programming Practice – beginner programs and problem-solving exercises.
- Python Learning Projects – small programs that demonstrate programming fundamentals.
- Networking / Cybersecurity Notes – beginner-friendly learning notes and safe lab exercises.

Note: These are suggested portfolio sections, not claims that the repositories already exist.

CAREER ROADMAP & CONNECT

A practical learning plan for the journey towards cybersecurity.

CAREER ROADMAP

Stage	Learning Focus	Portfolio Goal
Year 1 (2026–27)	C, Python, MySQL, computer and networking fundamentals	Publish programming practice and learning notes on GitHub
Year 2 (2027–28)	Operating systems, network protocols, Linux and security basics	Build beginner projects and document safe lab exercises
Year 3 (2028–29)	Network security, web security fundamentals, scripting and security tools	Create well-documented cybersecurity learning projects
Year 4 (2029–30)	Advanced security concepts, practical projects and internship preparation	Complete a capstone project and prepare a professional portfolio

CERTIFICATIONS & LEARNING GOALS

The following are future learning targets, not certifications already earned:

- CompTIA Security+ – consider after building the required fundamentals.
- Cisco networking learning pathway / CCNA – explore networking concepts in depth.
- Beginner cybersecurity courses and guided, legal practice labs.

ACHIEVEMENTS & ACTIVITIES

- Developing programming skills through coursework and regular practice.
- Interested in participating in hackathons, technical events and collaborative learning activities.
- Building a public record of learning and projects through GitHub.

CURRENT LEARNING FOCUS

01	Strengthen C and Python programming fundamentals.
02	Build a clear understanding of computer networking.
03	Continue learning MySQL and database concepts.
04	Explore cybersecurity fundamentals through safe, beginner-level practice.

CONNECT WITH ME

GitHub: https://github.com/tlharshini123-source

I’m interested in connecting with fellow students and technology enthusiasts who enjoy learning about programming, networking and cybersecurity.